DBMS LAB — EXPERIMENT 07

Normalization: UNF to 1NF, 2NF, 3NF and BCNF, Lossless Join and Dependency Preservation

Note: These are expected/representative outputs based on the described SQL and sample data, not screenshots from a live MySQL run. Exact formatting, timings and EXPLAIN plans can vary by MySQL version.

Output 1: Unnormalized table (Sales_UNF)

After inserting the sample data, Sales_UNF holds three rows. Product_IDs, Product_Names and Quantities store several comma-separated values in one cell, so the table is not in 1NF.

Sale_ID	Sale_Date	Customer_ID	Customer_Name	Customer_City	Salesperson_ID	Salesperson_Name	Product_IDs	Product_Names	Quantities	Payment_Mode
1001	2026-10-01	101	Aarav Sharma	Delhi	501	Neha Gupta	P01,P02	Laptop,Mouse	1,2	UPI
1002	2026-10-02	102	Riya Verma	Lucknow	502	Rahul Singh	P02,P03	Mouse,Keyboard	1,1	CARD
1003	2026-10-03	103	Karan Mehta	Kanpur	501	Neha Gupta	P01,P04	Laptop,Headphones	1,1	CASH

Output 2: Functional dependencies and candidate key

Determinant	Determines
Sale_ID	Sale_Date, Customer_ID, Salesperson_ID, Payment_Mode
Customer_ID	Customer_Name, Customer_City
Salesperson_ID	Salesperson_Name
Product_ID	Product_Name, Product_Category, Unit_Price
(Sale_ID, Product_ID)	Quantity
Candidate key in 1NF	(Sale_ID, Product_ID)

Output 3: Query 1 – Verify 1NF records

Each product now has its own row (3 sales expand to 6 rows). The primary key is (Sale_ID, Product_ID). Customer, salesperson and product details are still repeated, which causes partial and transitive dependencies.

Sale_ID	Sale_Date	Cust_ID	Customer_Name	Customer_City	Sp_ID	Salesperson_Name	Product_ID	Product_Name	Category	Unit_Price	Qty	Payment
1001	2026-10-01	101	Aarav Sharma	Delhi	501	Neha Gupta	P01	Laptop	Electronics	60000.00	1	UPI
1001	2026-10-01	101	Aarav Sharma	Delhi	501	Neha Gupta	P02	Mouse	Accessories	800.00	2	UPI
1002	2026-10-02	102	Riya Verma	Lucknow	502	Rahul Singh	P02	Mouse	Accessories	800.00	1	CARD
1002	2026-10-02	102	Riya Verma	Lucknow	502	Rahul Singh	P03	Keyboard	Accessories	1500.00	1	CARD
1003	2026-10-03	103	Karan Mehta	Kanpur	501	Neha Gupta	P01	Laptop	Electronics	60000.00	1	CASH
1003	2026-10-03	103	Karan Mehta	Kanpur	501	Neha Gupta	P04	Headphones	Accessories	2500.00	1	CASH

Output 4: Query 2 – Verify 2NF

Partial dependencies are removed by splitting into Sale_2NF (3 rows, key Sale_ID) and Sale_Item_2NF (6 rows, key Sale_ID + Product_ID). The join returns all 6 item rows:

Sale_ID	Customer_Name	Product_ID	Product_Name	Quantity
1001	Aarav Sharma	P01	Laptop	1
1001	Aarav Sharma	P02	Mouse	2
1002	Riya Verma	P02	Mouse	1
1002	Riya Verma	P03	Keyboard	1
1003	Karan Mehta	P01	Laptop	1
1003	Karan Mehta	P04	Headphones	1

Output 5: Query 3 – Verify 3NF

Transitive dependencies are removed. Row counts after decomposition: Customer 3, Salesperson 2, Product 4, Sale 3, Sale_Item 6. The five-table join reproduces the original 6 records:

Sale_ID	Sale_Date	Customer_Name	Salesperson_Name	Product_Name	Quantity	Payment_Mode
1001	2026-10-01	Aarav Sharma	Neha Gupta	Laptop	1	UPI
1001	2026-10-01	Aarav Sharma	Neha Gupta	Mouse	2	UPI
1002	2026-10-02	Riya Verma	Rahul Singh	Mouse	1	CARD
1002	2026-10-02	Riya Verma	Rahul Singh	Keyboard	1	CARD
1003	2026-10-03	Karan Mehta	Neha Gupta	Laptop	1	CASH
1003	2026-10-03	Karan Mehta	Neha Gupta	Headphones	1	CASH

Output 6: BCNF verification and SHOW CREATE TABLE

Every determinant in the final relations is a key of its own relation, so all five relations are in BCNF. SHOW CREATE TABLE returns the stored definition of each table (ENGINE=InnoDB, utf8mb4) with these keys:

Relation	Functional dependency	Determinant is superkey?	Primary key	Foreign keys
Customer	Customer_ID -> Customer_Name, Customer_City	Yes	Customer_ID	None
Salesperson	Salesperson_ID -> Salesperson_Name	Yes	Salesperson_ID	None
Product	Product_ID -> Product_Name, Product_Category, Unit_Price	Yes	Product_ID	None
Sale	Sale_ID -> Sale_Date, Customer_ID, Salesperson_ID, Payment_Mode	Yes	Sale_ID	Customer_ID -> Customer; Salesperson_ID -> Salesperson
Sale_Item	(Sale_ID, Product_ID) -> Quantity	Yes	(Sale_ID, Product_ID)	Sale_ID -> Sale; Product_ID -> Product

Output 7: Lossless join verification

The row count of the original 1NF table equals the row count of the join of the decomposed tables, so no rows are lost or invented.

Original_Rows	Reconstructed_Rows
6	6

Output 8: Dependency preservation verification

Zero violations means every functional dependency is still enforced inside the decomposed relations.

Check_Name	Violations
Customer FD violation	0
Salesperson FD violation	0
Product FD violation	0
Sale FD violation	0
Sale_Item key violation	0

Important

These are expected behaviors, not literal server-captured outputs. Row counts, timings, costs and plan wording must be checked against a live run of the exact SQL statements in your repository.